

LAB Logbook

Lab 1

Code :

```
a = np.arange(31)
```

```
print(a)
```

```
a1 = a.reshape(1, 31)
```

```
print(a1)
```

```
a2 = a1
```

```
print(a2)
```

```
print(a2.shape)
```

Output :

```
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
 24 25 26 27 28 29 30]
[[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
 24 25 26 27 28 29 30]]
[[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
 24 25 26 27 28 29 30]]
(1, 31)
```

The vector contains 31 elements because the last two digits of my SID are 31. After reshaping, the array has 1 row and 31 columns, giving a shape of (1, 31).

Lab 2

Code :

```
# 1. Determine n from the last digit of the SID=2520731
```

n = 1

2. Group by relationship and hours-per-week

```
Group_by_relationship_hours = data.groupby(["relationship", "hours-per week"])
```

```
Group_by_relationship_hours.size()
```

3. Reduce hours-per-week by n using a function

```
def reduce_hours(x):
```

```
    return x - n
```

```
data["hours-per-week-reduced"] = data["hours-per-week"].apply(reduce_hours)
```

```
data
```

4. Group by relationship and reduced hours-per-week

```
Group_by_relationship_reduced_hours = data.groupby(
```

```
    ["relationship", "hours-per-week-reduced"]
```

```
)
```

```
Group_by_relationship_reduced_hours.size()
```

Output :

...		0
relationship	hours-per-week-reduced	
Husband	12.0	1
	39.0	2
	44.0	1
	79.0	1
Not-in-family	15.0	1
	39.0	2
	49.0	2
Own-child	29.0	1
Wife	39.0	2
dtype: int64		

Lab 3

Lab 4

Lab 5

Lab 6

Lab 7

Lab 8

Lab 9

Lab 10

Lab 11

Lab 12